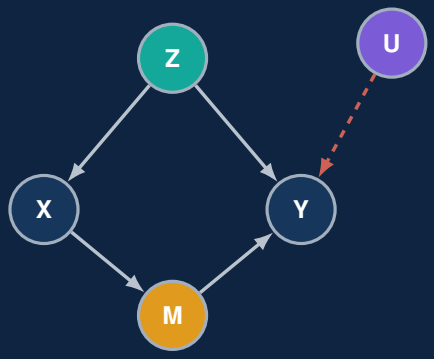


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 5

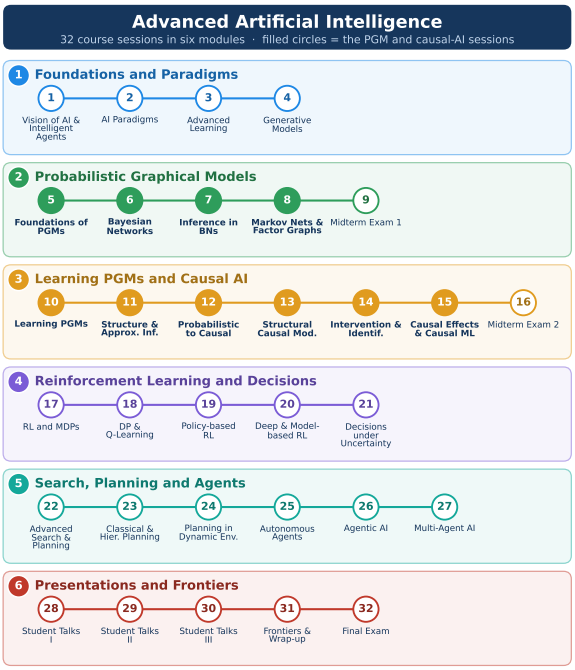
Foundations: Conditional Independence

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 5 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, the core theory organized in parts, worked numerical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s05_examples.py` for the numbers, `python/s05_figures.py` for the figures), and `questions/s05_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions; this is session 5.

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SESSION 5 ADVANCED ARTIFICIAL INTELLIGENCE

Foundations: Conditional Independence

Probability review, conditional independence, graphical representation, DAGs and factorization

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Probability Review
Axioms, Bayes, total probability, expectation, tower rule and total variance.

PART II
Conditional Independence
Definition, factorized form, graphoid axioms, worked examples.

PART III
Graphs & Factorization
From CI statements to DAGs, topological order, parameter counting, I-maps.

PART IV
AI Connections
Belief states, the Markov property, routers, multi-agent fusion.

LEARNING OBJECTIVES

- Apply Bayes' theorem, total probability, the tower property and the law of total variance.
- Define (conditional) independence, prove its factorized form and the graphoid axioms, and explain why intersection needs positivity.
- Turn CI statements into a DAG, factorize along a topological order and count parameters.
- Recognize CI behind belief states, the Markov property of RL, naive-Bayes routers and multi-agent fusion.

Where we are. Course sessions 1–4 introduced rational agents, the symbolic/statistical/neural paradigms, generalization and deep generative models — all of which manipulate high-dimensional distributions. This session opens the PGM block; its organizing idea is *conditional independence* (CI), which turns an intractable joint distribution into a product of small, learnable pieces. Session 6 turns CI statements into Bayesian networks, Sessions 7–11 compute with and learn them, and Sessions 12–15 give the arrows a causal meaning.

OPENING MOTIVATION

Suppose you want to build a full probabilistic model of $n = 50$ binary symptoms and diseases for a diagnostic system. The joint distribution $P(X_1, \ldots, X_{50})$ has $2^{50} - 1 \approx 1.13 \times 10^{15}$ free parameters — more than any dataset could estimate. Yet doctors reason about hundreds of symptoms every day, because most symptoms are *irrelevant to each other once the diagnosis is*

known. This session makes that intuition precise: conditional independence turns intractable joint distributions into tractable, learnable models, and every remaining session builds on it.

PART I

Probability Review

The language every later session speaks

PART I · PROBABILITY

5.1 Probability Review

5.1.1 Probability spaces and random variables

A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$. The *sample space* Ω collects all outcomes of an experiment; the *event space* $\mathcal{F}$ is a σ-algebra on Ω, i.e. a collection of subsets containing Ω and closed under complements and countable unions; and $\mathbb{P} : \mathcal{F} \to [0,1]$ satisfies Kolmogorov's axioms:

$$\mathbb{P}(A) \geq 0, \qquad \mathbb{P}(\Omega) = 1, \qquad \mathbb{P}\Big(\bigcup_{k} A_k\Big) = \sum_{k} \mathbb{P}(A_k) \ \text{ for pairwise disjoint } A_1, A_2, \ldots \tag{5.1}$$

A *random variable* is a measurable map $X : \Omega \to \mathcal{X}$; its distribution is $P_X(B) = \mathbb{P}(X^{-1}(B))$. We write $P(x)$ for $\mathbb{P}(X = x)$ when X is discrete (a probability mass function, pmf) and $p(x)$ for a density when X is continuous. For several variables, the *joint* distribution $P(x, y)$ determines the *marginals* $P(x) = \sum_y P(x, y)$ and, whenever $P(y) > 0$, the *conditionals*

$$P(x \mid y) = \frac{P(x, y)}{P(y)}. \tag{5.2}$$

Theorem 5.1 (Product rule, total probability and Bayes)**.** *For discrete random variables with the relevant probabilities positive:*

(i) ***Chain rule:*** $P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid x_1, \ldots, x_{i-1})$.

(ii) ***Law of total probability:*** $P(y) = \sum_x P(y \mid x) P(x)$.

(iii) ***Bayes' theorem:*** $P(x \mid y) = \dfrac{P(y \mid x) P(x)}{\sum_{x'} P(y \mid x') P(x')}$.

The same statements hold for densities with sums replaced by integrals.

Proof. (i) Apply (5.2) repeatedly: $P(x_1, \ldots, x_n) = P(x_n \mid x_{<n}) P(x_{<n})$ and induct on n. (ii) The events $\{X = x\}$ partition Ω, so additivity in (5.1) gives $P(y) = \sum_x P(x, y) = \sum_x P(y \mid x) P(x)$. (iii) Write $P(x \mid y) = P(x, y)/P(y)$, expand the numerator by the product rule and the denominator by (ii). □

Worked Example 5.1 (Bayes' theorem for a screening test)**.** *Data (hypothetical).* $D \in \{0, 1\}$ indicates disease (D=1) or no disease (D=0); "+" and "−" denote a positive and a negative

test result. The test has *sensitivity* $P(+ \mid D{=}1) = 0.95$ (probability that a diseased person tests positive), *specificity* $P(- \mid D{=}0) = 0.90$ (probability that a healthy person tests negative), and the *prevalence* is $P(D{=}1) = 0.02$. Find $P(D{=}1 \mid +)$, and then $P(D{=}1 \mid +,+)$ after a second, independent repetition of the test.

Solution.

Step 1. Complements. The theorem needs $P(+ \mid D{=}0)$, which is not given directly; it is the complement of the specificity (the *false-positive rate*): $P(+ \mid D{=}0) = 1 - P(- \mid D{=}0) = 1 - 0.90 = 0.10$. Likewise $P(D{=}0) = 1 - 0.02 = 0.98$.

Step 2. Evidence by total probability (Theorem 5.1(ii)):

$$\begin{aligned} P(+) &= P(+ \mid D{=}1)P(D{=}1) + P(+ \mid D{=}0)P(D{=}0) \\ &= 0.95 \cdot 0.02 + 0.10 \cdot 0.98 = 0.019 + 0.098 = 0.117. \end{aligned}$$

Step 3. Posterior by Bayes (Theorem 5.1(iii)):

$$P(D{=}1 \mid +) = \frac{P(+ \mid D{=}1)P(D{=}1)}{P(+)} = \frac{0.019}{0.117} = 0.1624.$$

Of the 0.117 probability mass of positives, 0.098 comes from healthy people and only 0.019 from diseased ones: roughly one positive in six ($1/0.1624 \approx 6.2$) is a true case, because the 98% healthy majority produces most positives.

Step 4. A second positive test. Assume the two results T_1, T_2 are *conditionally independent given* D, $T_1 \perp\!\!\!\perp T_2 \mid D$ (Definition 5.2 below). Then the likelihoods multiply: $P(+,+ \mid D{=}1) = 0.95^2 = 0.9025$ and $P(+,+ \mid D{=}0) = 0.10^2 = 0.0100$. Hence

$$\begin{aligned} P(D{=}1 \mid +,+) &= \frac{P(+,+ \mid D{=}1)P(D{=}1)}{P(+,+ \mid D{=}1)P(D{=}1) + P(+,+ \mid D{=}0)P(D{=}0)} \\ &= \frac{0.9025 \cdot 0.02}{0.9025 \cdot 0.02 + 0.0100 \cdot 0.98} = \frac{0.01805}{0.01805 + 0.00980} \\ &= \frac{0.01805}{0.02785} = 0.6481. \end{aligned}$$

(The same value follows by using 0.1624 as the prior for the second test.)

Without the CI assumption we would need the joint table $P(T_1, T_2 \mid d)$ (three free numbers per value of d) instead of the single number $P(+ \mid d)$ — our first encounter with CI as a modelling assumption that makes computation possible.

PYTHON LAB · s05_examples.py · ex_bayes_review()

```
P(+|D=0) = 1 - 0.9 = 0.10   P(+) = 0.019 + 0.098 = 0.1170   P(D=1|+) =
0.0190/0.1170 = 0.1624
P(D=1|+,+) = 0.01805/(0.01805 + 0.00980) = 0.6481   sequential:
0.1543/0.2380 = 0.6481
```

5.1.2 Expectation, conditional expectation and the tower property

The expectation of $g(X)$ is $\mathbb{E}[g(X)] = \sum_x g(x)P(x)$ (or $\int g(x)p(x)\,dx$). The *conditional expectation* $\mathbb{E}[Y \mid X]$ is the random variable $m(X)$ where $m(x) = \mathbb{E}[Y \mid X{=}x] = \sum_y y\,P(y \mid x)$; it is the best predictor of Y from X in mean square. The conditional variance is $\mathrm{Var}(Y \mid X) = \mathbb{E}[(Y - \mathbb{E}[Y \mid$

$X])^2 \mid X]$.

Proposition 5.1 (Tower property and law of total variance)**.** *For square-integrable Y:*

$$\mathbb{E}[Y] = \mathbb{E}\big[\mathbb{E}[Y \mid X]\big], \qquad \operatorname{Var}(Y) = \underbrace{\mathbb{E}\big[\operatorname{Var}(Y \mid X)\big]}_{\textit{within-group}} + \underbrace{\operatorname{Var}\big(\mathbb{E}[Y \mid X]\big)}_{\textit{between-group}}.$$

Proof. Tower (discrete case): $\mathbb{E}[\mathbb{E}[Y \mid X]] = \sum_x P(x) \sum_y y\, P(y \mid x) = \sum_{x,y} y\, P(x,y) = \mathbb{E}[Y]$. For the variance write $m(X) = \mathbb{E}[Y \mid X]$ and $Y - \mathbb{E}Y = (Y - m(X)) + (m(X) - \mathbb{E}Y)$. Squaring and taking expectations, $\operatorname{Var}(Y) = \mathbb{E}[(Y-m)^2] + \mathbb{E}[(m - \mathbb{E}Y)^2] + 2\mathbb{E}[(Y-m)(m-\mathbb{E}Y)]$. By the tower property the first term is $\mathbb{E}[\operatorname{Var}(Y \mid X)]$, the second is $\operatorname{Var}(m(X))$ since $\mathbb{E}[m(X)] = \mathbb{E}Y$, and the cross term is $2\mathbb{E}\big[(m(X) - \mathbb{E}Y)\,\mathbb{E}[Y - m(X) \mid X]\big] = 0$. □

PART I · PROBABILITY

5.2 Why the Joint Distribution Alone Is Not Enough

For discrete random variables $X_1, \ldots, X_n$, the chain rule of probability (Theorem 5.1) lets us write *any* joint distribution, with no assumptions at all, as

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P\big(x_i \mid x_1, \ldots, x_{i-1}\big). \tag{5.3}$$

This is always true, and it is almost always useless: the i-th factor is a table indexed by all $i-1$ preceding variables, so the total number of free parameters is still $2^n - 1$ for binary variables. The chain rule tells us a joint distribution *can* be factored; it says nothing about *when the factors are small*. Figure 5.1 makes the blow-up concrete: with only ten binary variables, a full joint table already needs $2^{10} - 1 = 1023$ parameters, while the chain $X_1 \to X_2 \to \cdots \to X_{10}$, in which each variable depends only on its predecessor, needs 1 number for $P(X_1{=}1)$ plus 2 numbers ($P(X_i{=}1 \mid x_{i-1})$ for $x_{i-1} = 0, 1$) for each of the other nine variables: $1 + 2 \cdot 9 = 19$.

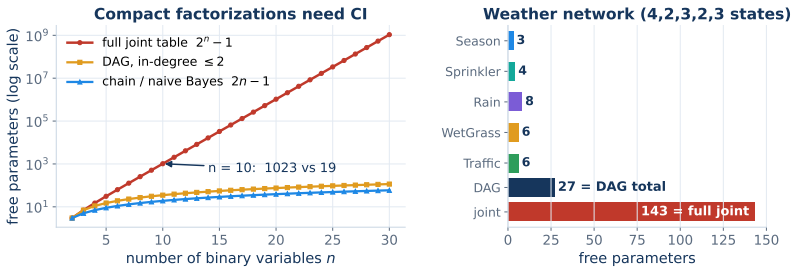


Figure 5.1. Left: free parameters required to specify a joint distribution over n binary variables (log scale): the full joint table ($2^n - 1$), a DAG in which every variable from the third on has two parents ($1 + 2 + 4(n-2)$), and a chain or naive-Bayes model ($1 + 2(n-1) = 2n - 1$). Right: the mixed-cardinality weather network of Worked Example 5.6 needs $3 + 4 + 8 + 6 + 6 = 27$ parameters instead of 143.

The gap between these curves is entirely explained by conditional independence (CI) assumptions. This is the organizing idea of the whole PGM block: *a graphical model is nothing more than a compact way of writing down which conditional independencies hold, and a computationally efficient way of using them.* Everything from Session 6 onward is machinery built on top of this one idea.

PART II

Conditional Independence

Definition, factorization and the algebra of irrelevance

PART II · INDEPENDENCE

5.3 Independence and Conditional Independence

Definition 5.1 (Independence)**.** Random variables X and Y are *independent*, written $X \perp\!\!\!\perp Y$, if

$$P(x, y) = P(x)P(y) \quad \text{for all values } x, y,$$

equivalently $P(x \mid y) = P(x)$ whenever $P(y) > 0$: learning Y tells you nothing about X.

Independence is rare in practice — almost everything is at least weakly related to everything else. The far more useful and far more common structure is independence *given some context*.

Definition 5.2 (Conditional Independence)**.** Let X, Y, Z be (possibly vector-valued) random variables. X and Y are *conditionally independent given* Z, written $X \perp\!\!\!\perp Y \mid Z$, if

$$P(x, y \mid z) = P(x \mid z)\, P(y \mid z) \quad \text{for all } x, y, z \text{ with } P(z) > 0.$$

Equivalently, $P(x \mid y, z) = P(x \mid z)$ whenever $P(y, z) > 0$. For sets of variables, $\mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z}$ means the statement holds for the random vectors. Marginal independence is the special case $Z = \emptyset$.

Notation

CI gap (used in the Python checks). Multiplying the defining equation by $P(z)^2$ shows that $X \perp\!\!\!\perp Y \mid Z$ holds iff $P(x, y, z)P(z) = P(x, z)P(y, z)$ for all x, y, z (cells with $P(z) = 0$ satisfy it trivially). The function `ci_gap` in `s05_examples.py` returns $\Delta(X; Y \mid Z) = \max_{x,y,z} \left|P(x, y, z)P(z) - P(x, z)P(y, z)\right|$, which is 0 exactly when CI holds; for $Z = \emptyset$ it is $\max_{x,y} |P(x, y) - P(x)P(y)|$.

Figure 5.2 pictures this with overlapping regions of shared information: under $X \perp\!\!\!\perp Y \mid Z$ the *entire* overlap of X and Y sits inside Z, so once Z is known nothing of the X–Y relationship is left to learn.

An equivalent and often more convenient characterization avoids explicitly normalizing:

Proposition 5.2 (Factorized form of CI)**.** *$X \perp\!\!\!\perp Y \mid Z$ if and only if there exist functions g and h such that*

$$P(x, y, z) = g(x, z)\, h(y, z) \quad \textit{for all } x, y, z.$$

Proof. ($\Rightarrow$) Take $g(x, z) = P(x \mid z)P(z)$ and $h(y, z) = P(y \mid z)$; then $g(x, z)h(y, z) = P(x \mid z)P(y \mid z)P(z) = P(x, y \mid z)P(z) = P(x, y, z)$ by Definition 5.2. ($\Leftarrow$) With $G(z) = \sum_x g(x, z)$ and $H(z) = \sum_y h(y, z)$, summing $P(x, y, z) = g(x, z)h(y, z)$ over x, over y and over both gives $P(y, z) =$

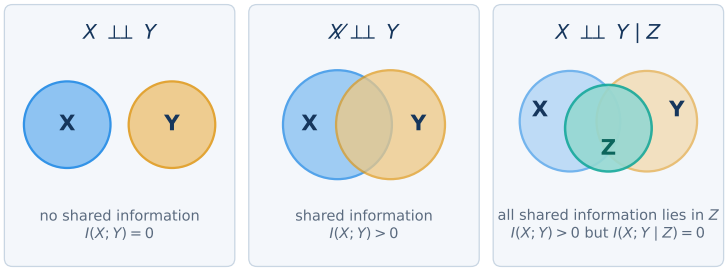


Figure 5.2. Information-diagram view. Left: independence, no shared information. Middle: dependence. Right: $X \perp\!\!\!\perp Y \mid Z$ — all shared information between X and Y routes through Z, so X and Y are dependent marginally but independent once Z is known.

$G(z)h(y,z)$, $P(x,z) = g(x,z)H(z)$ and $P(z) = G(z)H(z)$. Hence $P(x,y \mid z) = \frac{g(x,z)h(y,z)}{G(z)H(z)} = \frac{g(x,z)H(z)}{P(z)} \cdot \frac{G(z)h(y,z)}{P(z)} = P(x \mid z)P(y \mid z)$. □

This factorized view is exactly why CI matters computationally: it says the three-way table $P(x,y,z)$ can be built from two smaller tables g and h instead of one large one — a preview of the factor-based computation in Session 7. It is also the template for the DAG factorization of Part III (Theorem 5.2).

COMMON PITFALL Neither $X \perp\!\!\!\perp Y$ nor $X \perp\!\!\!\perp Y \mid Z$ implies the other. A common cause can create marginal dependence that disappears given Z (Worked Example 5.4); conditioning on a common effect can create dependence that was absent marginally (Worked Example 5.5).

PART II · INDEPENDENCE

5.4 The Algebra of Conditional Independence: Graphoid Axioms

CI is not an arbitrary relation; it obeys algebraic rules that let us manipulate independence statements symbolically, the way we manipulate equations.

Proposition 5.3 (Properties of conditional independence)**.** *For disjoint sets of variables X, Y, Z, W:*

(i) ***Symmetry:*** $X \perp\!\!\!\perp Y \mid Z \iff Y \perp\!\!\!\perp X \mid Z$.

(ii) ***Decomposition:*** $X \perp\!\!\!\perp (Y,W) \mid Z \implies X \perp\!\!\!\perp Y \mid Z$.

(iii) ***Weak union:*** $X \perp\!\!\!\perp (Y,W) \mid Z \implies X \perp\!\!\!\perp Y \mid (Z,W)$.

Proof. Symmetry: multiplication commutes. Decomposition: sum $P(x,y,w \mid z) = P(x \mid z)P(y,w \mid z)$ over w. Weak union: divide the same identity by $P(w \mid z) > 0$ to get $P(x,y \mid z,w) = P(x \mid z)P(y \mid z,w)$; summing over y gives $P(x \mid z,w) = P(x \mid z)$, hence $P(x,y \mid z,w) = P(x \mid z,w)P(y \mid z,w)$. □

In words: decomposition says irrelevance of a group implies irrelevance of any subgroup; weak union says X stays irrelevant to Y after we additionally condition on W. Session 6's d-separation

reads such consequences directly off a picture. Two further rules complete the system.

Proposition 5.4 (Contraction and intersection)**.** *For disjoint X, Y, Z, W:*

(iv) ***Contraction:*** *$X \perp\!\!\!\perp W \mid Z$ and $X \perp\!\!\!\perp Y \mid (Z, W) \implies X \perp\!\!\!\perp (Y, W) \mid Z$.*

(v) ***Intersection:*** *if P is strictly positive, $X \perp\!\!\!\perp Y \mid (Z, W)$ and $X \perp\!\!\!\perp W \mid (Z, Y) \implies X \perp\!\!\!\perp (Y, W) \mid Z$.*

Proof. Contraction: by the two hypotheses $P(x \mid y, w, z) = P(x \mid w, z) = P(x \mid z)$ wherever defined, so $P(x, y, w \mid z) = P(x \mid y, w, z)P(y, w \mid z) = P(x \mid z)P(y, w \mid z)$. Intersection: the hypotheses give $P(x \mid y, w, z) = P(x \mid w, z)$ and $P(x \mid y, w, z) = P(x \mid y, z)$, so $P(x \mid w, z) = P(x \mid y, z)$ for all x, y, w, z. The left side does not depend on y and the right side not on w; since $P > 0$, every combination (y, w) occurs, so both equal some $f(x, z)$. Summing $P(x \mid y, z)P(y \mid z) = f(x, z)P(y \mid z)$ over y gives $f(x, z) = P(x \mid z)$, hence $P(x \mid y, w, z) = P(x \mid z)$. □

A relation satisfying (i)–(iv) is called a *semi-graphoid*; with (v) as well it is a *graphoid* (Pearl 2009; Koller & Friedman 2009, Sec. 2.1). Every probability distribution induces a semi-graphoid; strictly positive distributions induce graphoids. Graph separation also satisfies all five rules, which is the deep reason why graphs are a good language for CI.

Worked Example 5.2 (Intersection fails without positivity)**.** Let $X = Y = W$ be three copies of one fair bit, so $P(X{=}0, Y{=}0, W{=}0) = P(X{=}1, Y{=}1, W{=}1) = \frac{1}{2}$ and the other six cells are 0 (take $Z = \emptyset$). *Hypotheses hold.* Given $W = w$, both X and Y equal w, so $P(x, y \mid w) = \mathbb{1}\{x = w\}\mathbb{1}\{y = w\} = P(x \mid w)P(y \mid w)$ for all x, y: $X \perp\!\!\!\perp Y \mid W$. By the same argument with the roles of Y and W exchanged, $X \perp\!\!\!\perp W \mid Y$. *Conclusion fails.* $X \perp\!\!\!\perp (Y, W)$ would require $P(X{=}1, Y{=}1, W{=}1) = P(X{=}1)P(Y{=}1, W{=}1)$, but the left side is $\frac{1}{2}$ and the right side is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$; the CI gap is $\frac{1}{2} - \frac{1}{4} = 0.25$. In fact X determines (Y, W) completely. Intersection fails because the zero cells (e.g. $Y = 0, W = 1$) prevent us from "moving" between values of W while holding Y fixed. Deterministic relations — common in logic-based AI and in software agents — are exactly where positivity breaks.

PYTHON LAB · s05_examples.py · ex_intersection_counterexample()

```
gap X indep Y | W : 0.0   gap X indep W | Y : 0.0   gap X indep (Y,W) : 0.25
```

PART II · INDEPENDENCE

5.5 Worked Examples

Worked Example 5.3 (Symptoms given a diagnosis)**.** A clinic models three symptoms — fever S_1, cough S_2, fatigue S_3 — and a binary diagnosis $D \in \{\text{flu}, \text{no flu}\}$. Suppose the modeler assumes each symptom is generated independently once the diagnosis is fixed:

$$S_1 \perp\!\!\!\perp S_2 \perp\!\!\!\perp S_3 \mid D$$

(shorthand for every pair being CI given D, and jointly $P(s_1, s_2, s_3 \mid d) = \prod_{k=1}^{3} P(s_k \mid d)$). This is the classical *naive Bayes* assumption. Without it, modeling $P(s_1, s_2, s_3 \mid d)$ requires $2^3 - 1 = 7$ parameters per value of d; with it, only 3 parameters per value of d are needed

— one number $P(S_k{=}1 \mid d)$ per symptom. Including the prior $P(D{=}\text{flu})$, the model needs $1+2\cdot 3=7$ numbers instead of $1+2\cdot 7=15$. The assumption is almost certainly not perfectly true (a cough may make fatigue more likely even after conditioning on flu status), but it is often close enough to be highly useful, and it is learnable from a small dataset where the exact joint would not be.

Worked Example 5.4 (Marginal vs. conditional independence can point opposite ways)**.** Consider a recommendation system with Z = "user is a movie enthusiast" (binary), X = "user rated Movie A highly," Y = "user rated Movie B highly." Suppose that, given enthusiast status, ratings for the two unrelated movies are independent: $X \perp\!\!\!\perp Y \mid Z$, with $P(X{=}1 \mid Z{=}1)=0.9$, $P(X{=}1 \mid Z{=}0)=0.1$, and identically for Y; enthusiasts make up 30% of users, so $P(Z{=}1)=0.3$. Show that X and Y are *marginally dependent* despite being conditionally independent.
Solution. By the law of total probability,

$$P(X{=}1,Y{=}1)=\sum_z P(X{=}1 \mid z)P(Y{=}1 \mid z)P(z)=(0.9)^2(0.3)+(0.1)^2(0.7)$$
$$=0.81\cdot 0.3+0.01\cdot 0.7=0.243+0.007=0.250,$$

where the first equality uses $X \perp\!\!\!\perp Y \mid Z$. Marginally $P(X{=}1)=0.9(0.3)+0.1(0.7)=0.27+0.07=0.34$ (and the same for Y), so if X,Y were independent we would need $P(X{=}1,Y{=}1)=0.34^2=0.1156$, far from the true value 0.250. Equivalently, $\mathrm{Cov}(X,Y)=P(X{=}1,Y{=}1)-P(X{=}1)P(Y{=}1)=0.250-0.1156=0.1344>0$, and learning $X=1$ raises the probability of $Y=1$ from 0.34 to $P(Y{=}1 \mid X{=}1)=0.250/0.34=0.7353$. The unobserved enthusiasm variable Z *induces* an apparent correlation between two otherwise unrelated ratings — a small preview of the confounding phenomena that will become central in Session 12.

PYTHON LAB · s05_examples.py · ex_enthusiast()

```
P(X=1,Y=1) = 0.243 + 0.007 = 0.250   P(X=1) = 0.27 + 0.07 = 0.34; P(X=1)^2 =
0.1156
Cov(X,Y) = 0.1344   P(Y=1 | X=1) = 0.25/0.34 = 0.7353
```

Worked Example 5.5 (Pairwise independence does not imply mutual independence — and flips under conditioning)**.** Let X,Y be independent fair coin flips, and let $Z=X\oplus Y$ (addition mod 2). We show all three pairs are *marginally* independent, yet $X \not\perp\!\!\!\perp Y \mid Z$.
Marginal independence of every pair. By symmetry $P(X{=}1)=P(Y{=}1)=P(Z{=}1)=\frac{1}{2}$. Now $P(X{=}1,Z{=}1)=P(X{=}1,Y{=}0)=\frac{1}{4}=P(X{=}1)P(Z{=}1)$, so $X \perp\!\!\!\perp Z$; by an identical computation $Y \perp\!\!\!\perp Z$; and $X \perp\!\!\!\perp Y$ was assumed. So X,Y,Z are *pairwise* independent.
But not mutually independent. If X,Y,Z were mutually independent, we would need $P(X{=}1,Y{=}1,Z{=}1)=P(X{=}1)P(Y{=}1)P(Z{=}1)=\frac{1}{8}$. But $Z=X\oplus Y$, so $X{=}1,Y{=}1$ forces $Z{=}0$ with certainty: $P(X{=}1,Y{=}1,Z{=}1)=0\neq\frac{1}{8}$.
Conditioning on Z destroys the pairwise independence of X,Y. Given $Z=0$, we know $Y=X$ exactly (since $X\oplus Y=0 \iff X=Y$), so $P(X{=}1 \mid Y{=}1,Z{=}0)=1\neq P(X{=}1 \mid Z{=}0)=\frac{1}{2}$: knowing Y (given Z) tells you X with certainty. Hence $X \not\perp\!\!\!\perp Y \mid Z$, even though $X \perp\!\!\!\perp Y$ unconditionally. In terms of the CI gap: in the cell $(x,y,z)=(1,1,0)$, $P(1,1,0)P(Z{=}0)=\frac{1}{4}\cdot\frac{1}{2}=\frac{1}{8}$ while $P(X{=}1,Z{=}0)P(Y{=}1,Z{=}0)=\frac{1}{4}\cdot\frac{1}{4}=\frac{1}{16}$, a gap of $\frac{1}{16}=0.0625$. Together with Worked Example 5.4 (CI given a common cause, yet marginal dependence), this shows that neither of $X \perp\!\!\!\perp Y$ and $X \perp\!\!\!\perp Y \mid Z$ implies the other.

PYTHON LAB · s05_examples.py · ex_xor_pairwise()

```
pairwise CI gaps 0.0, 0.0, 0.0   P(X=1,Y=1,Z=1) = 0.0   gap X indep Y | Z =
0.0625
```

PART III

Graphical Representation: DAGs and Factorization

From lists of CI statements to pictures

PART III · GRAPHS

5.6 DAG Vocabulary

Definition 5.3 (Directed acyclic graph)**.** A *DAG* $G = (V, E)$ is a finite set of nodes $V = \{X_1, \ldots, X_n\}$ with directed edges $E \subseteq V \times V$ and no directed cycle. For a node X: its *parents* $\mathrm{pa}(X) = \{U : U \to X\}$; *children* $\mathrm{ch}(X) = \{U : X \to U\}$; *ancestors* $\mathrm{an}(X)$: nodes with a directed path to X; *descendants* $\mathrm{de}(X)$: nodes reachable from X by a directed path; *non-descendants* $\mathrm{nd}(X) = V \setminus (\{X\} \cup \mathrm{de}(X))$. A *topological order* is an ordering of V in which every parent precedes its children.

Every DAG has at least one topological order: a finite DAG has a node without parents (otherwise following parent edges backwards would revisit a node and close a cycle); put it first, delete it and recurse. Parents are always non-descendants, $\mathrm{pa}(X) \subseteq \mathrm{nd}(X)$, since otherwise a directed cycle would exist.

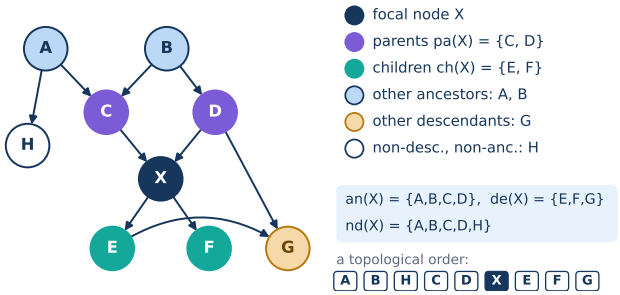


Figure 5.3. DAG vocabulary relative to a focal node X, and one topological order (there are several: e.g. H may be placed anywhere after A).

PART III · GRAPHS

5.7 From CI Statements to a DAG

Pick an ordering $X_1, \ldots, X_n$ of the variables. Suppose that for each i we can find a (small) subset $\mathrm{pa}(X_i) \subseteq \{X_1, \ldots, X_{i-1}\}$ such that

$$X_i \perp\!\!\!\perp \{X_1, \ldots, X_{i-1}\} \setminus \mathrm{pa}(X_i) \mid \mathrm{pa}(X_i). \tag{5.4}$$

Then each factor of the chain rule (5.3) simplifies, $P(x_i \mid x_1, \dots, x_{i-1}) = P(x_i \mid \mathrm{pa}(x_i))$, and drawing an arrow from each member of $\mathrm{pa}(X_i)$ to X_i produces a DAG. This is the whole recipe for a Bayesian network; Session 6 develops it in full.

Definition 5.4 (Local Markov property; I-map)**.** Let $\mathcal{I}(P)$ be the set of CI statements that hold in P. A DAG G satisfies the *local Markov property* for P if

$$\mathcal{I}_\ell(G) = \big\{\, X \perp\!\!\!\perp \mathrm{nd}(X) \setminus \mathrm{pa}(X) \mid \mathrm{pa}(X) \;:\; X \in V \big\} \;\subseteq\; \mathcal{I}(P).$$

We then call G an *I-map* (independence map) of P. It is a *minimal* I-map if removing any edge destroys this property.

Theorem 5.2 (Factorization along a topological order)**.** *Let G be a DAG over $X_1, \dots, X_n$. Then G is an I-map of P if and only if P factorizes over G:*

$$P(x_1, \dots, x_n) = \prod_{i=1}^{n} P\big(x_i \mid \mathrm{pa}(x_i)\big). \tag{5.5}$$

Proof. ($\Rightarrow$) Relabel so that $X_1, \dots, X_n$ is a topological order (it exists, see above). The predecessors $\{X_1, \dots, X_{i-1}\}$ contain $\mathrm{pa}(X_i)$ and are non-descendants of X_i. The local Markov property gives $X_i \perp\!\!\!\perp \mathrm{nd}(X_i) \setminus \mathrm{pa}(X_i) \mid \mathrm{pa}(X_i)$, and by decomposition (Proposition 5.3) $X_i \perp\!\!\!\perp \{X_1, \dots, X_{i-1}\} \setminus \mathrm{pa}(X_i) \mid \mathrm{pa}(X_i)$, i.e. (5.4). Substituting into the chain rule (5.3) gives (5.5). ($\Leftarrow$) Fix X_i, let $N = \mathrm{nd}(X_i)$, and sum (5.5) over the descendants of X_i, each time eliminating one with no remaining children (its CPT sums to one). This leaves $P(x_i, x_N) = P(x_i \mid \mathrm{pa}(x_i)) \prod_{j \in N} P(x_j \mid \mathrm{pa}(x_j))$, because parents of non-descendants are non-descendants: a function of $(x_i, \mathrm{pa}(x_i))$ times a function of $x_N \supseteq \mathrm{pa}(x_i)$. By Proposition 5.2 with $Z = \mathrm{pa}(X_i)$, $X_i \perp\!\!\!\perp N \setminus \mathrm{pa}(X_i) \mid \mathrm{pa}(X_i)$. □

5.7.1 Parameter counting

If X_i has k_i states, the CPT $P(X_i \mid \mathrm{pa}(X_i))$ has one probability vector (with $k_i - 1$ free entries) for each of the $\prod_{j \in \mathrm{pa}(i)} k_j$ parent configurations. Hence

$$\#\mathrm{params}(G) = \sum_{i=1}^{n} (k_i - 1) \prod_{j \in \mathrm{pa}(i)} k_j \qquad \text{versus} \qquad \prod_{i=1}^{n} k_i - 1 \;\text{ for the full joint.} \tag{5.6}$$

For binary variables and in-degree at most d the BN needs at most $n 2^d$ numbers: linear rather than exponential in n.

Worked Example 5.6 (Counting parameters with mixed cardinalities)**.** A hypothetical weather network has Season (4 states) $\to$ Sprinkler (2), Season $\to$ Rain (3: none/light/heavy), (Sprinkler, Rain) $\to$ WetGrass (2), Rain $\to$ Traffic (3: light/medium/heavy). Formula (5.6) gives, node by node: Season $(4-1) \cdot 1 = 3$; Sprinkler $(2-1) \cdot 4 = 4$; Rain $(3-1) \cdot 4 = 8$; WetGrass $(2-1) \cdot (2 \cdot 3) = 6$; Traffic $(3-1) \cdot 3 = 6$. In total $3 + 4 + 8 + 6 + 6 = 27$ free parameters, while the full joint over $4 \cdot 2 \cdot 3 \cdot 2 \cdot 3 = 144$ configurations needs $144 - 1 = 143$ — a factor of $143/27 = 5.30$ even for five variables.

PYTHON LAB · s05_examples.py · ex_param_count()

```
Season (4-1)x1=3, Sprinkler (2-1)x4=4, Rain (3-1)x4=8, WetGrass (2-1)x6=6,
Traffic (3-1)x3=6
BN total = 27   full joint = 144 - 1 = 143   ratio 5.30
```

5.7.2 A first look at I-maps: the ordering matters

For a fixed joint P, the recipe (5.4) with the *smallest* admissible parent sets yields a minimal I-map, but which I-map we get depends on the ordering. Under positivity the smallest parent set for a given ordering is unique (a consequence of weak union, intersection, decomposition and contraction; Pearl 1988, *Probabilistic Reasoning in Intelligent Systems*, Sec. 3.3).

Worked Example 5.7 (Minimal I-maps of the sprinkler distribution for all 24 orderings)**.** Take the binary sprinkler network (Cloudy, Sprinkler, Rain, WetGrass) $C \to S$, $C \to R$, $(S, R) \to W$ with illustrative CPTs $P(C{=}1) = 0.5$; $P(S{=}1 \mid C{=}0) = 0.5$, $P(S{=}1 \mid C{=}1) = 0.1$; $P(R{=}1 \mid C{=}0) = 0.2$, $P(R{=}1 \mid C{=}1) = 0.8$; $P(W{=}1 \mid s, r) = 0.05, 0.9, 0.9, 0.99$ for $(s, r) = (0,0), (1,0), (0,1), (1,1)$. For every ordering, `ex_imap_orders()` computes the joint by (5.5) and gives each X_i the smallest set U of predecessors with CI gap $\Delta\big(X_i; \text{predecessors} \setminus U \mid U\big) < 10^{-10}$, i.e. the smallest parent set satisfying (5.4); a binary node with |pa| binary parents costs $2^{|\text{pa}|}$ parameters.

- **Causal ordering** C, S, R, W recovers the generating graph: 4 edges and $1+2+2+4 = 9$ parameters (for C, S, R, W respectively).
- **Ordering** W, R, S, C gives $\text{pa}(R) = \{W\}$, $\text{pa}(S) = \{R, W\}$, $\text{pa}(C) = \{S, R\}$: 5 edges and $1 + 2 + 4 + 4 = 11$ parameters. S needs *both* predecessors because $S \not\perp\!\!\!\perp R \mid W$ (CI gap 0.0684) and $S \not\perp\!\!\!\perp W \mid R$ (gap 0.0518); the first is "explaining away" (Session 6): conditioning on the common effect W couples its causes.
- **Ordering** W, C, S, R even yields a complete DAG: $\text{pa}(C) = \{W\}$, $\text{pa}(S) = \{C, W\}$, $\text{pa}(R) = \{C, S, W\}$, i.e. 6 edges and $1 + 2 + 4 + 8 = 15$ parameters — no saving at all over the full joint ($2^4 - 1 = 15$).

Across all 24 orderings: 4 give 9 parameters, 8 give 11 and 12 give 15.

PYTHON LAB · s05_examples.py · ex_imap_orders()

```
order CSRW: params 9, edges 4   order WRSC: params 11, edges 5   order WCSR:
params 15, edges 6
counts over 24 orderings:  {9:  4, 11:  8, 15:  12}   gaps:  S|R given W
0.0684, S|W given R 0.0518
S|R given C 0.00e+00, S|R given C,W 0.0076, C|W given S,R 4.34e-19
```

Every distribution has many I-maps. Sparse ones are found by ordering causes before effects — the first hint that causal knowledge (Sessions 12–15) makes graphical models compact.

PART IV

Conditional Independence in AI, Agents and RL

Belief states, Markov property, routers and multi-agent fusion

PART IV · CONNECTIONS

5.8 Connections to AI, Agentic AI, RL and Multi-Agent Systems

5.8.1 Agentic AI: belief states and naive-Bayes routing

AI CONNECTION · Agentic AI

An LLM assistant does not observe the user's intent U; it observes utterances $O_1, O_2, \ldots$. If utterances are modelled as CI given the intent, $O_t \perp\!\!\!\perp O_{1:t-1} \mid U$, the agent's *belief state* $b_t(u) = P(U{=}u \mid o_{1:t})$ satisfies the recursion

$$b_t(u) \;\propto\; P(o_t \mid u)\, b_{t-1}(u),$$

one multiplication per turn, with memory independent of conversation length (when users refer back to earlier turns the CI fails and the latent state must be enriched — the POMDP view below; course session 26). As a *naive-Bayes router* the agent calls the tool (billing API, troubleshooter, CRM) of the most probable intent.

Worked Example 5.8 (Tracking a hidden user intent)**.** Intents $u \in \{\text{Billing, TechSupport, Sales}\}$ with prior $b_0 = (0.5, 0.3, 0.2)$ (all vectors below are in this order); hypothetical likelihoods $P(\text{“invoice”} \mid u) = (0.6, 0.1, 0.2)$ and $P(\text{“error”} \mid u) = (0.2, 0.7, 0.05)$ for the words the user types in turns 1 and 2. Each update multiplies the current belief by the likelihood entrywise and divides by the sum (the evidence).

- After “invoice”: unnormalized $(0.5 \cdot 0.6,\ 0.3 \cdot 0.1,\ 0.2 \cdot 0.2) = (0.30, 0.03, 0.04)$, evidence $0.30 + 0.03 + 0.04 = 0.37$, belief $b_1 = (0.30, 0.03, 0.04)/0.37 = (0.8108, 0.0811, 0.1081)$.
- After “error” (CI given u lets us reuse b_1 as the prior): unnormalized $(0.8108 \cdot 0.2,\ 0.0811 \cdot 0.7,\ 0.1081 \cdot 0.05) = (0.1622, 0.0568, 0.0054)$, evidence 0.2243, belief $b_2 = (0.7229, 0.2530, 0.0241)$.

Billing still wins, but TechSupport has more than tripled ($0.0811 \to 0.2530$): a clarifying question may be worth its cost (value of information, course session 21).

PYTHON LAB · `s05_examples.py` · `ex_agent_intent()`

```
after 'invoice':  unnormalised (0.3000, 0.0300, 0.0400), evidence 0.3700,
belief (0.8108, 0.0811, 0.1081)
after 'error':    unnormalised (0.1622, 0.0568, 0.0054), evidence 0.2243,
belief (0.7229, 0.2530, 0.0241)
```

5.8.2 Reinforcement learning: the Markov property

AI CONNECTION · Reinforcement Learning

An MDP is defined by a CI statement. With history $H_{t-1} = (S_0, A_0, \ldots, S_{t-1}, A_{t-1})$, the *Markov property* reads

$$S_{t+1} \perp\!\!\!\perp H_{t-1} \mid S_t, A_t, \qquad \text{equivalently} \qquad P(s_{t+1} \mid s_{0:t}, a_{0:t}) = P(s_{t+1} \mid s_t, a_t).$$

It makes the transition model a single table $P(s' \mid s, a)$. In a POMDP the state is hidden and observations satisfy $O_t \perp\!\!\!\perp$ (everything else) $\mid S_t$; the belief state $b_t(s) = P(S_t{=}s \mid o_{1:t}, a_{0:t-1})$ is itself Markov, with update $b_{t+1}(s') \propto P(o_{t+1} \mid s') \sum_s P(s' \mid s, a_t) b_t(s)$, which is why POMDPs reduce to MDPs over beliefs (course session 21).

LINK TO COURSE SESSION 17 · RL and MDPs

Policies and values are defined relative to a state with $S_{t+1}, R_{t+1} \perp\!\!\!\perp H_{t-1} \mid S_t, A_t$. The Bellman equation $V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a)[r + \gamma V^\pi(s')]$ is the chain rule (5.3) along time, simplified by exactly this CI statement.

5.8.3 Multi-agent AI: private signals and sensor fusion

AI CONNECTION · Multi-Agent AI

Agents $1, \ldots, m$ observe private noisy signals $Y_1, \ldots, Y_m$ of a common state S (a hazard, a price, an opponent's type), assumed $Y_i \perp\!\!\!\perp Y_j \mid S$. Then the posterior log-odds is a *sum* of individual log-likelihood ratios,

$$\log \frac{P(S{=}1 \mid y_{1:m})}{P(S{=}0 \mid y_{1:m})} = \log \frac{P(S{=}1)}{P(S{=}0)} + \sum_{i=1}^{m} \log \frac{P(y_i \mid S{=}1)}{P(y_i \mid S{=}0)},$$

so each agent need only send one number. If CI fails (a relay, shared data, gossip), naive fusion double-counts evidence and pooled beliefs become over-confident (course session 27).

5.8.4 Generative models and AI paradigms

LINK TO COURSE SESSION 4 · Generative Models

A VAE decoder factorizes $p_\theta(x \mid z) = \prod_j p_\theta(x_j \mid z)$: pixels (or tokens) are CI given the latent code — the naive-Bayes structure with a continuous, learned z. All pixel dependence must be carried by z. Session 10 derives the ELBO from EM.

PART IV · CONNECTIONS

5.9 Summary

Table 5.1. Cheat-sheet of key formulas.

Result	Formula
Bayes / total probability	$P(x \mid y) = P(y \mid x)P(x) / \sum_{x'} P(y \mid x')P(x')$
Law of total variance	$\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X])$
CI definition	$P(x, y \mid z) = P(x \mid z)P(y \mid z)$
Factorized form	$X \perp\!\!\!\perp Y \mid Z \iff P(x, y, z) = g(x, z)h(y, z)$
DAG factorization	$P(x_{1:n}) = \prod_i P(x_i \mid \mathrm{pa}(x_i))$
Parameter count	$\sum_i (k_i - 1) \prod_{j \in \mathrm{pa}(i)} k_j$
Markov property	$S_{t+1} \perp\!\!\!\perp H_{t-1} \mid S_t, A_t$

THINK ABOUT IT **Which CI statements in your favourite ML model are assumptions, and which are guaranteed by its construction?**

Further reading. Koller & Friedman (2009), *Probabilistic Graphical Models*, Ch. 2–3. Pearl (1988), *Probabilistic Reasoning in Intelligent Systems*, Ch. 3 (graphoids, I-maps). Sutton & Barto (2018), Ch. 3 (Markov property).

CONNECTIONS AND TAKEAWAYS

This session established that (1) the chain rule always factors a joint distribution, but only CI makes the factors small; (2) CI has a product-factorization form (Proposition 5.2); (3) CI obeys the graphoid rules; (4) a list of CI statements yields a DAG and a compact factorization; and (5) CI is the assumption behind belief states, the Markov property and multi-agent fusion. What we do not yet have is a way to read off *which* CI statements hold without listing them by hand. **Session 6** answers this: a directed graph whose missing edges encode CI assumptions, and a graphical test (d-separation) that reads the implied CI statements off the picture.